Engagement, Ohio

Demographics and Relationships in a Mid-Sized American Town

Group 5 — Mini-Challenge 1

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Abstract

We present a coordinated visual analytics tool for exploring the demographics, social structure, and business base of Engagement, Ohio, the synthetic mid-sized American town described in the VAST Challenge 2022 dataset. Our analysis is restricted to Mini-Challenge 1 (demographics and relationships); the `ParticipantStatusLogs` (~5 GB of 5-minute location records) belong to MC2 and are deliberately not ingested, with one bounded exception used for cross-sectional financial summaries. The tool combines a 2D geographic map, a 3D wage-extruded heightmap coloured by joviality, and a four-panel coordinated dashboard (demographics, social network, business base, and per-employer detail) linked through a single shared selection. We answer the three core questions and present a one-page synthesis directed at city residents.

1 Q1 — Demographics of Engagement, Ohio

The study population comprises 1,011 participants across 1,042 buildings (residential, commercial, school). All findings below should be read with two caveats: the sample represents an unknown fraction of the city's true population, and 131 participants drop out of the collection after month one (§5).

Age structure. The age distribution is broadly uniform across the 18–60 range (Fig. 1). No single decade dominates; counts in each two-year bin sit between 38 and 70 with no clear modal peak. This indicates a stable, demographically balanced town rather than one skewed by either a recent youth influx or an aging population.

Education. The vast majority of residents hold at least a high-school diploma, and roughly 40% hold a bachelor's degree or higher (Fig. 2). Educational attainment is largely age-invariant, with two notable exceptions. The 18–24 cohort under-represents graduate degrees, consistent with that group not yet having had time to complete graduate education. The 25–34 cohort shows the highest graduate-degree share, suggesting that the most recent generation is pursuing higher education at greater rates than its predecessors.

Household composition. Approximately one-third of participants live alone, one-third in couples without children, and one-third in households with children. Group D — the highest-earning interest group — is the most family-oriented, with 43% of members reporting children in the household and the largest mean household size (2.16 people). Single-person households are concentrated in the central residential districts; family households are more spread to the city periphery (visible in the household composition tile of the demographics card, Fig. 3).

Wellbeing (joviality). Joviality is recorded once at study start as a continuous score in $[0, 1]$. The empirical city-wide mean is 0.49 with most values falling between 0.42 and 0.56 — a much narrower band than the theoretical range. We render joviality on the map (Fig. 5) using a sequential plasma colormap fitted to this empirical range rather than a divergent palette, because 0.5 has no semantic meaning for joviality and the white-midpoint of common divergent palettes collides with the cartographic "no-data" convention.

Spending. The average resident's monthly outgoings break down as: Shelter $680, Recreation $398, Food $345, Education $49 (Fig. 6). The Recreation/Food near-parity is notable: residents spend essentially as much on optional leisure as on necessary food, indicating a population with meaningful disposable income. Median rent is $707/month, broadly consistent with the shelter-spending figure. Education spending is low because the dataset covers a city with five schools serving children of resident parents; adult continuing-education expenditure is minimal.

Interest groups. Each participant self-reports one of ten interest groups (A–J), with specific topics redacted to avoid researcher bias. Group sizes are broadly balanced (83–116 members). Mean joviality varies modestly across groups: Group E is happiest (0.53) despite below-average education and wages; Group H is least happy (0.45) despite above-average income — a paradox we return to in Q2.

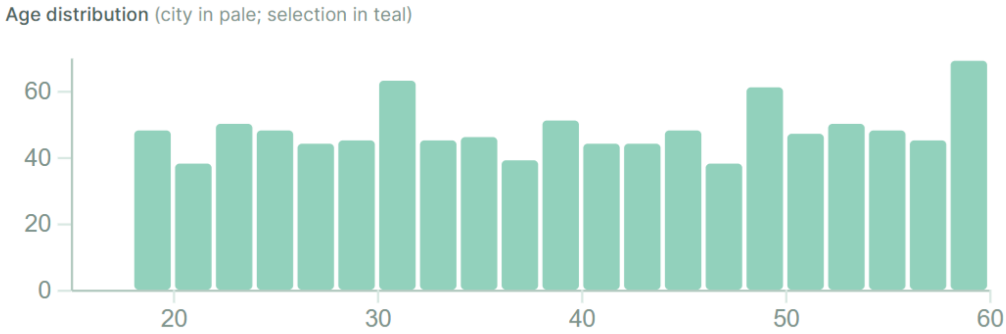

Figure 1: Age distribution of the 1,011 study participants. The distribution is uniform across the 18–60 range with no dominant cohort.

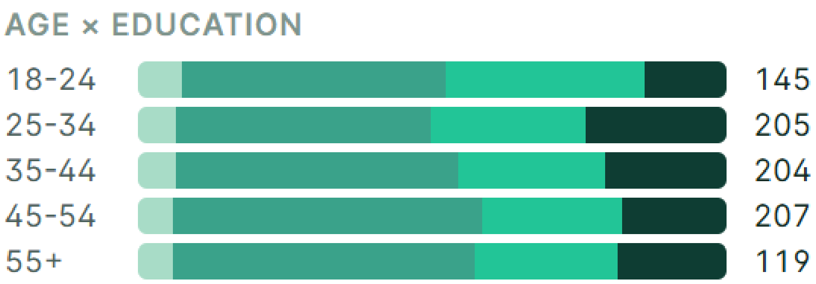

Figure 2: Education attainment by age band. Light to dark: Low, HighSchool/College, Bachelors, Graduate. The 25–34 cohort shows the highest graduate-degree share.

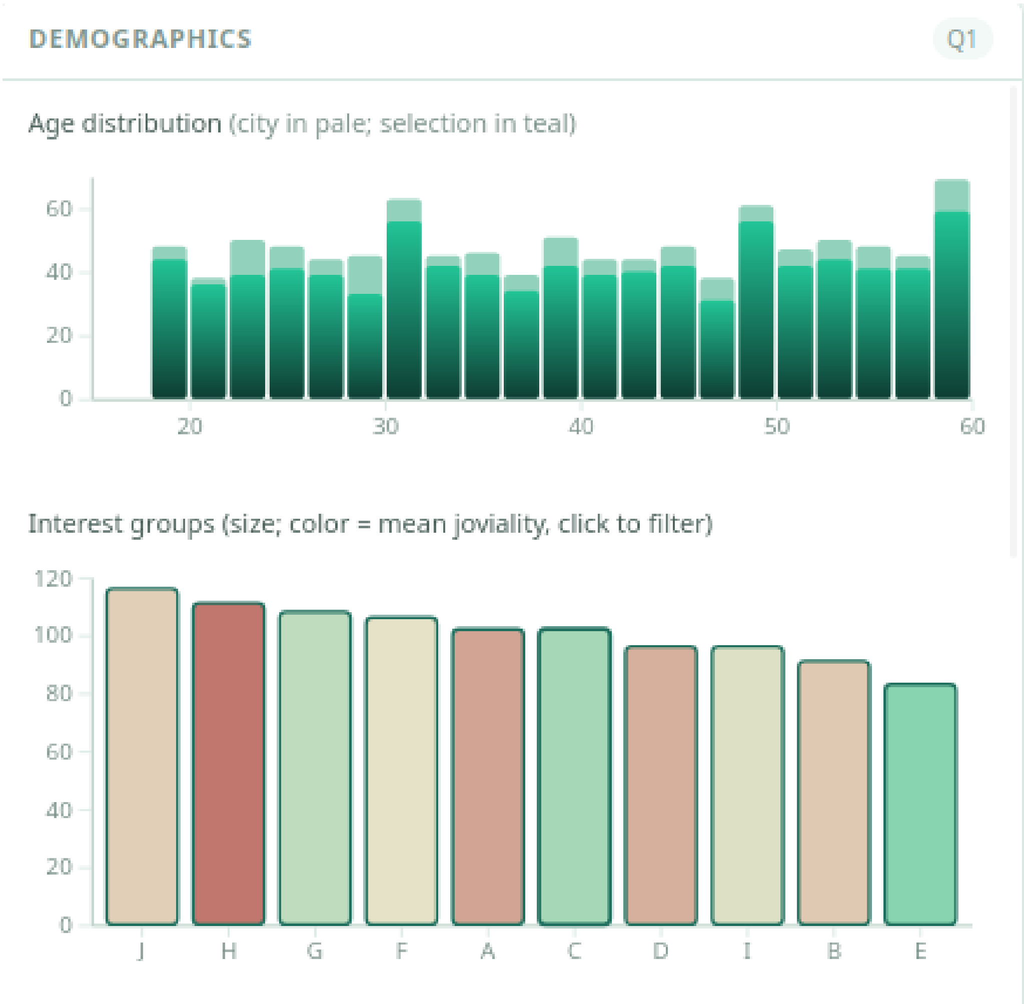

Figure 3: Demographics dashboard panel.

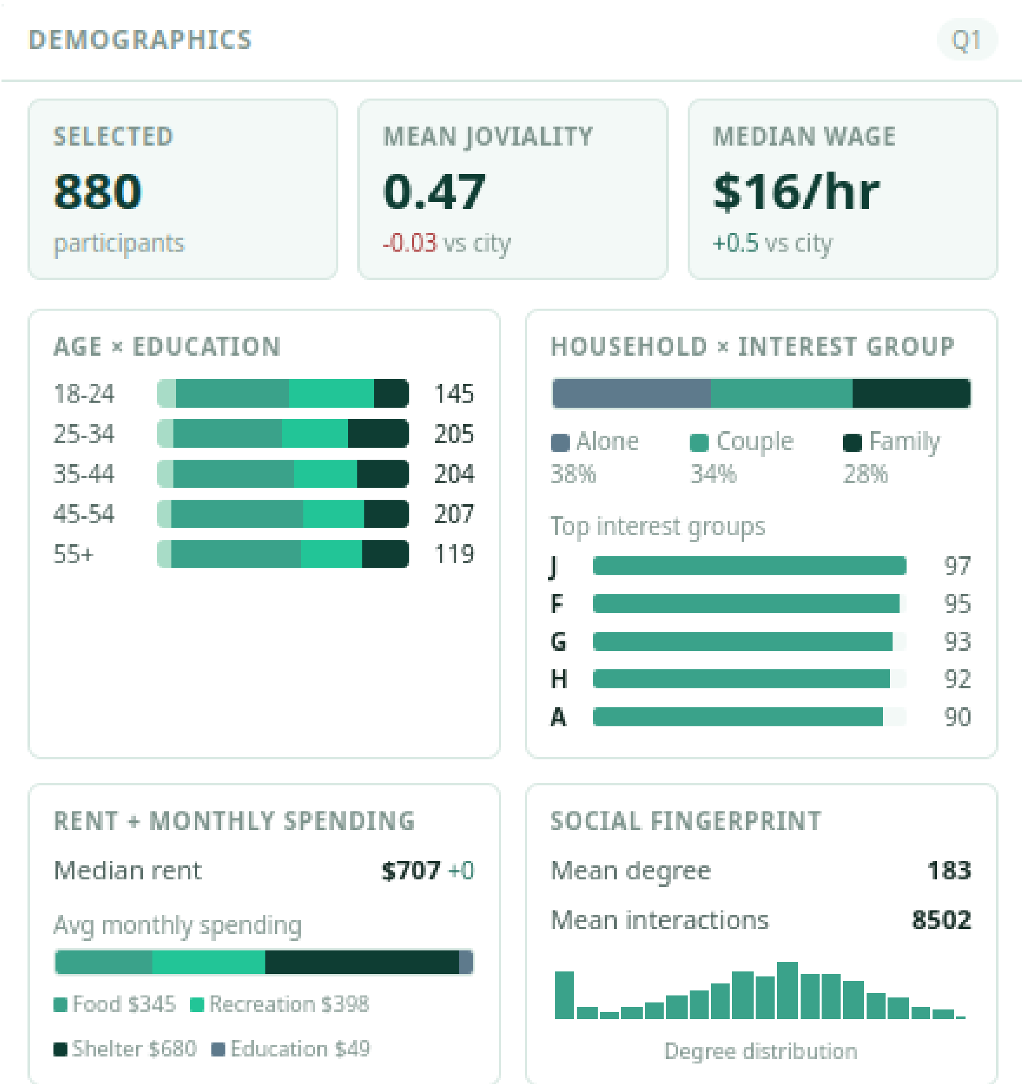

Figure 4: Demographics dashboard panel: KPIs and composition tiles summarise age, education, household, and spending at city scale.

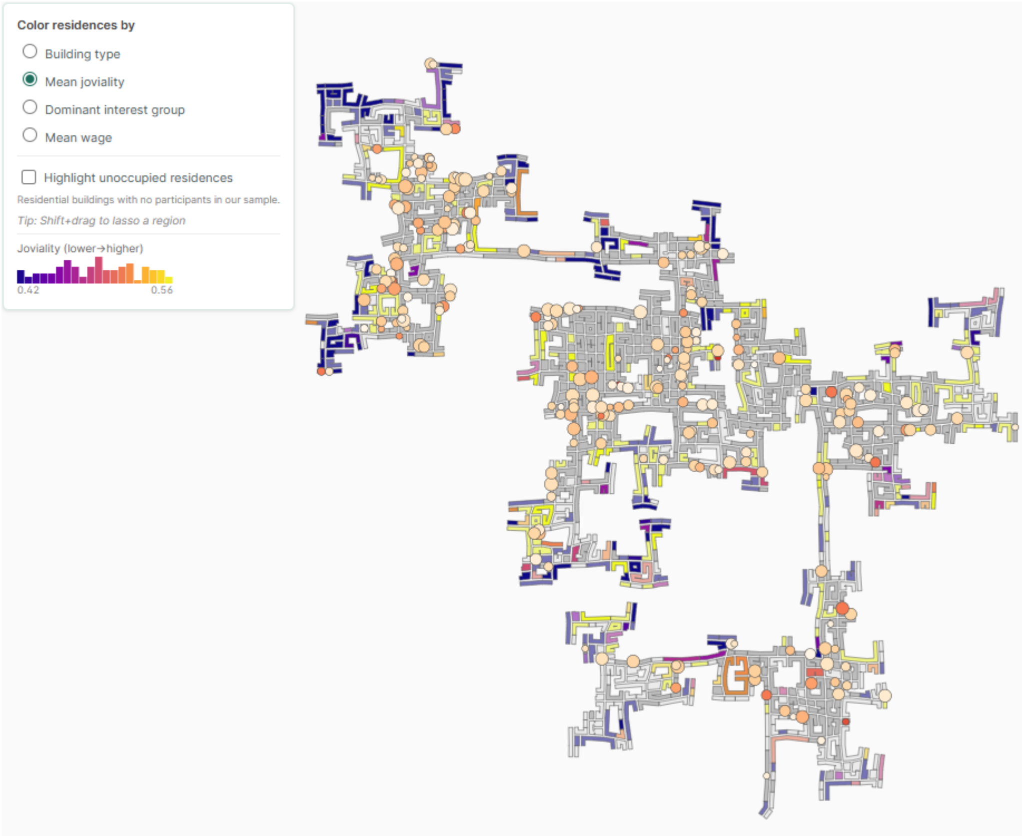

Figure 5: Mean joviality per residential building, sequential plasma colormap on the empirical range $[0.42, 0.56]$. The narrow empirical band reveals that joviality variation is more concentrated than the theoretical $[0, 1]$ scale suggests.

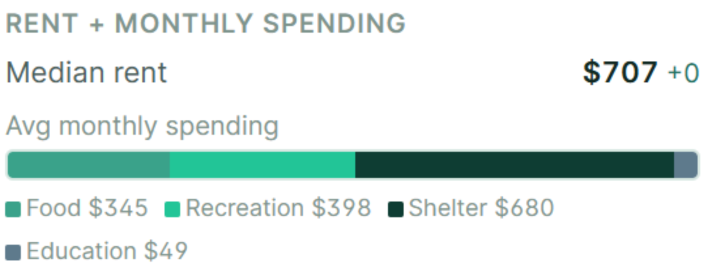

Figure 6: Average monthly spending breakdown. Shelter dominates; Recreation and Food are at near-parity, suggesting meaningful disposable income.

2 Q2 — Social Network Patterns

Scope note. The challenge brief's phrasing of "social activities" is ambiguous: it could imply behavioural data (when, where, how often people meet), which falls under MC2's "Patterns of Life". The patterns described here are derived from network *structure*, group membership, and joviality — the data sources cleanly within MC1. Where temporal behaviour is unavoidable (e.g. pub vs. restaurant weekend share), we flag it explicitly.

Pattern 1 — Intra-group interaction dominates. The group × group interaction matrix (Fig. 7) shows a clear diagonal: every group interacts most heavily with itself. Groups A, F, I, and J show the darkest self-interaction cells, indicating tight internal communities with strong homophily.

Pattern 2 — E↔F and G↔H form cross-group bridges. The strongest off-diagonal cells in the matrix are E↔F and G↔H, suggesting these pairs share meaningful ties beyond their own group. When Groups E and F are co-selected (189 participants), their connection lines span the full city and their buildings are geographically interspersed, indicating proximity drives this cross-group bridge (Fig. 8).

Pattern 3 — I and J form a tight social cluster. The I↔J cell is among the darker cross-group entries, and both groups show comparable mean interaction counts (768–847). Groups I and J appear socially adjacent, behaving almost as a super-group.

Pattern 4 — Friend count strongly predicts joviality. The friend-count vs. joviality scatter (Fig. 9) shows a clear positive trend. Residents with fewer than 50 friends concentrate in the low-joviality range (0.0–0.3); those with 200+ friends predominantly score above 0.6. Social connectivity is the strongest joviality correlate in the dataset.

Pattern 5 — The Group H paradox. Despite being the largest group (111 members) with the highest raw interaction count (707) and above-average income ($24.66/hr, $5,855 balance), Group H has the lowest mean joviality (0.45) and the fewest unique social connections (290 mean degree) (Fig. 10). High activity volume does not translate to wellbeing — quantity of social contact and quality of social experience appear decoupled.

Pattern 6 — Group E is the smallest and most insular, yet happiest. With only 83 members and the weakest cross-group matrix presence, Group E appears socially self-contained. Its mean joviality (0.53) is the highest of any group, suggesting that smaller, tighter communities may foster greater individual wellbeing.

Pattern 7 — Social networks are geographically unconstrained. Connection arcs visible across all group-filtered map views span the full city extent, with long-range ties crossing neighbourhood boundaries (Fig. 11). Friendships in Engagement are not locally bounded; social groups do not correspond to residential clusters.

Pattern 8 — Mild geographic clustering by interest group. The dominant-interest-group choropleth reveals loose spatial patterns (Group A concentrations in the north-west, I/J more prominent in the south) but no strict neighbourhood segregation. Most blocks show mixed group presence, consistent with Pattern 7.

Pattern 9 — Education and wage tier do not predict group membership. The Sankey diagram of Education → Interest Group → Wage tier (Fig. 12) shows ribbons from every education tier (Low 84, HS/College 525, Bachelors 232, Graduate 170) flowing into every interest group with broadly proportional widths. No group is skewed toward a particular education level or wage band ($10–15, $15–25, $25–40). A formal Kruskal-Wallis test on social connection count across groups confirms this is the only variable significantly differentiated by group membership ($H = 30.0$, $p = 0.0004$, $\eta^2 = 2.8\%$). All other tested variables — age, joviality, education, household size, having children, and financial measures — return $p > 0.05$ with effect sizes below 1.5% of variance.

Pattern 10 — Group size does not predict social cohesion. Comparing Groups F (106), G (108), and J (116) shows the largest groups do not have the highest mean interaction counts or joviality. Group G (joviality 0.51, mean interactions 793) outperforms larger Group J (0.49) on both metrics. Internal cohesion, not size, drives social activity.

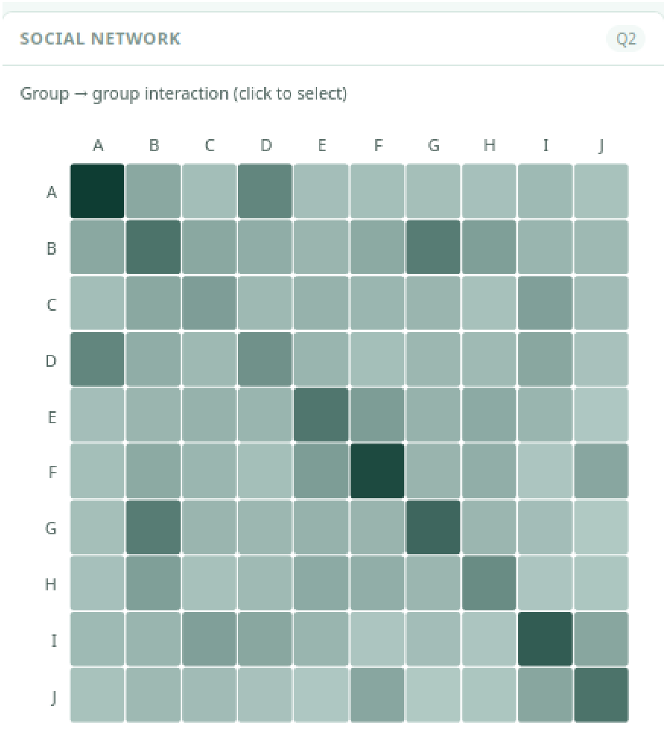

Figure 7: Group × group interaction matrix. Diagonal cells are consistently darkest, confirming intra-group ties dominate.

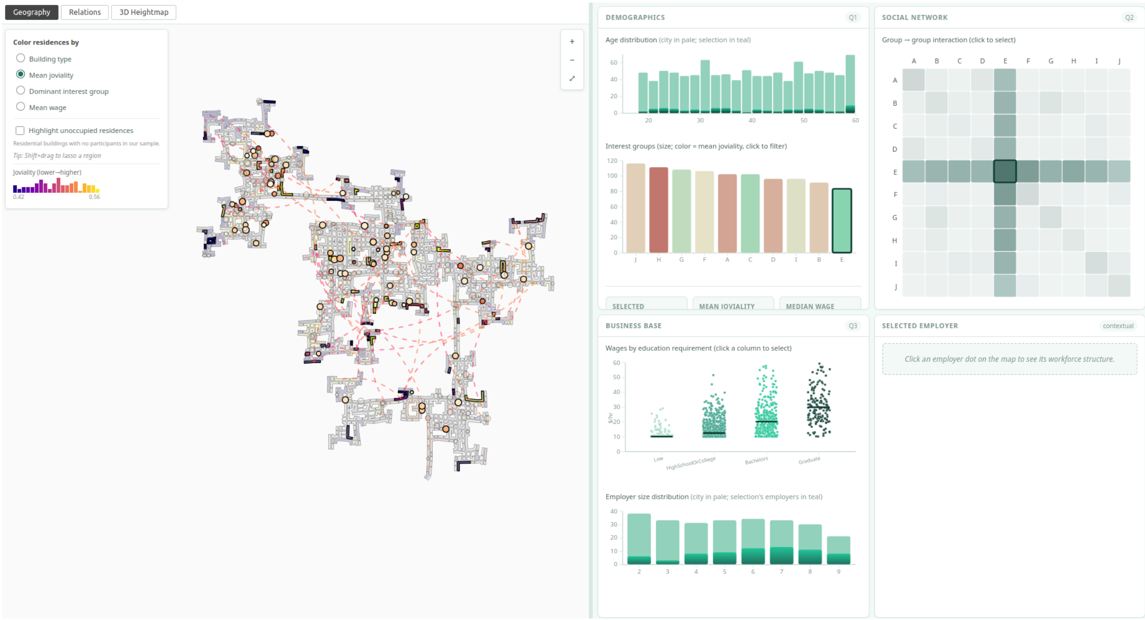

Figure 8: Group E selected (83 participants). Connection arcs span the full city, illustrating non-local friendship structure even within the smallest, most insular group.

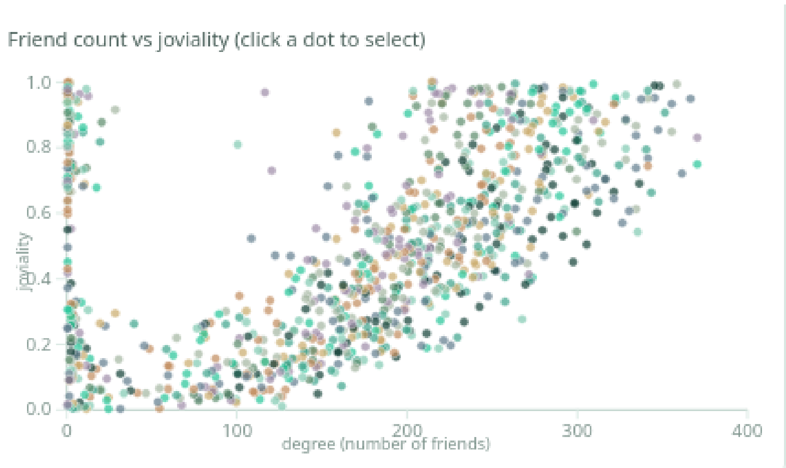

Figure 9: Friend count vs. joviality. Strong positive trend; isolated residents cluster bottom-left.

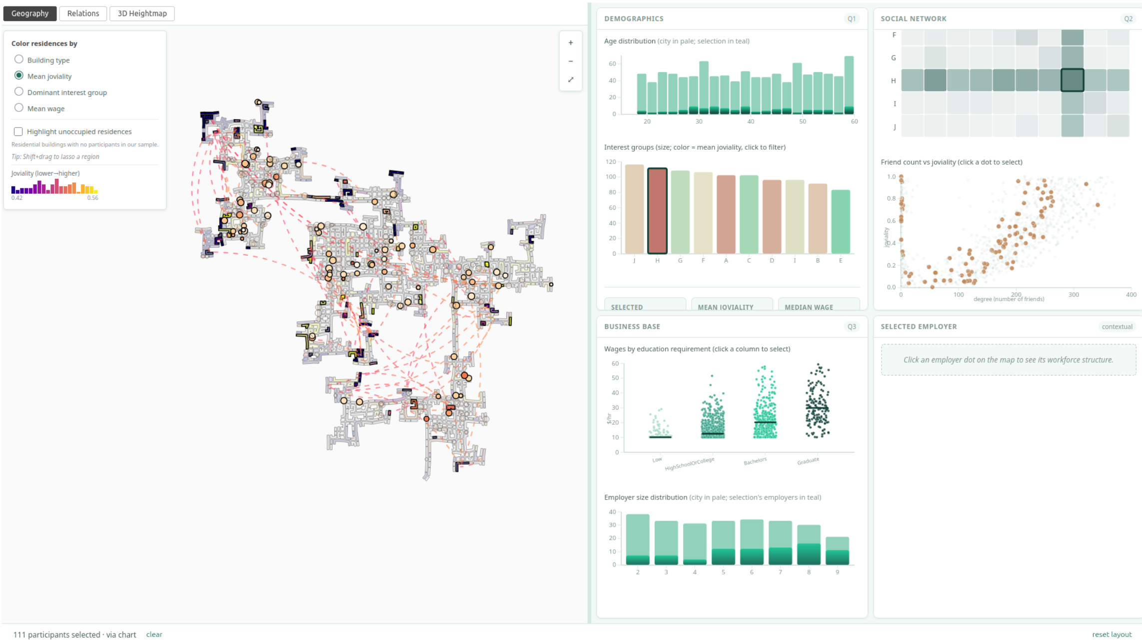

Figure 10: Group H selected (111 participants). Highest interaction volume yet lowest joviality and fewest unique connections.

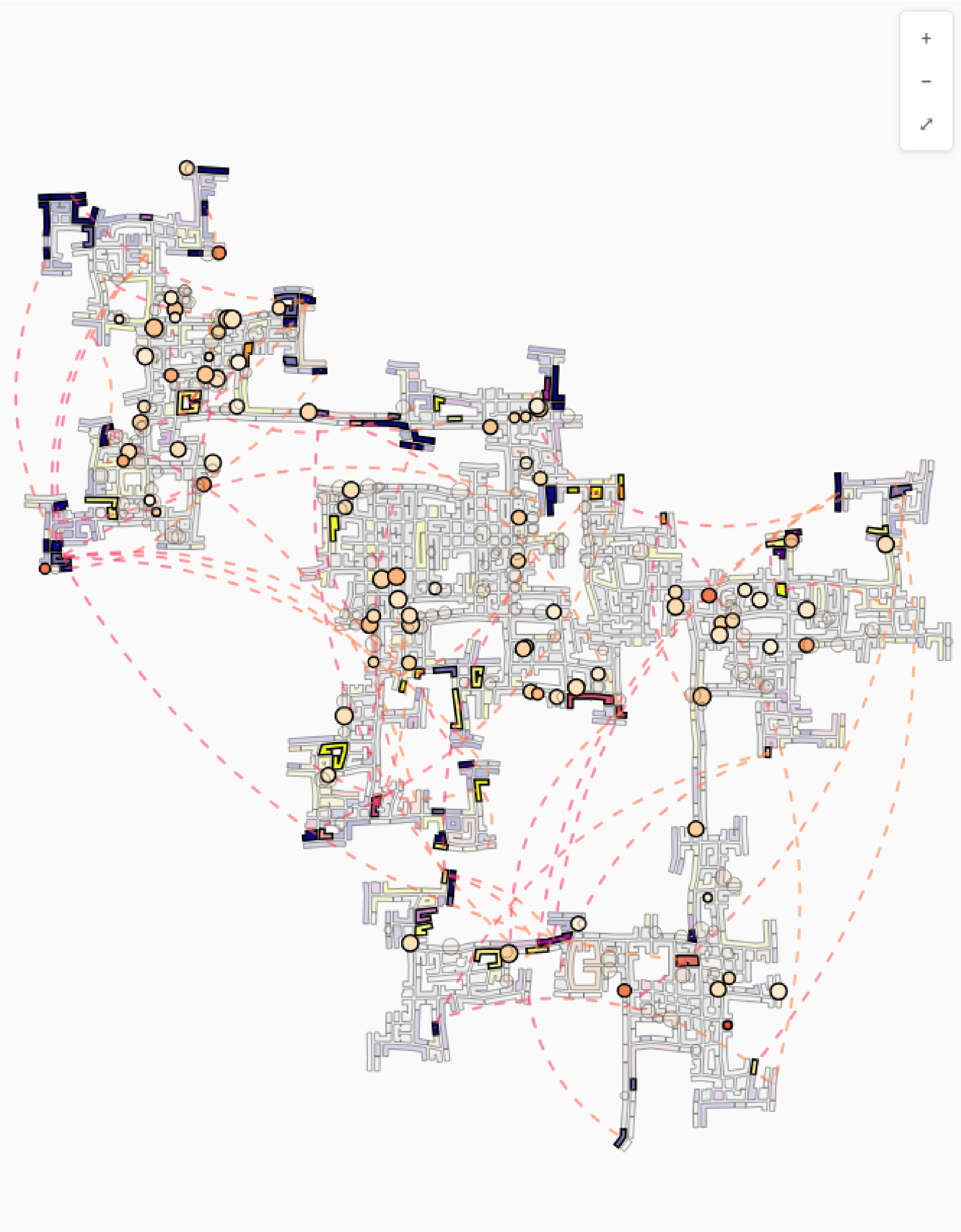

Figure 11: Group A selected (102 participants). Connection arcs span the entire city, indicating geographically distributed friendships.

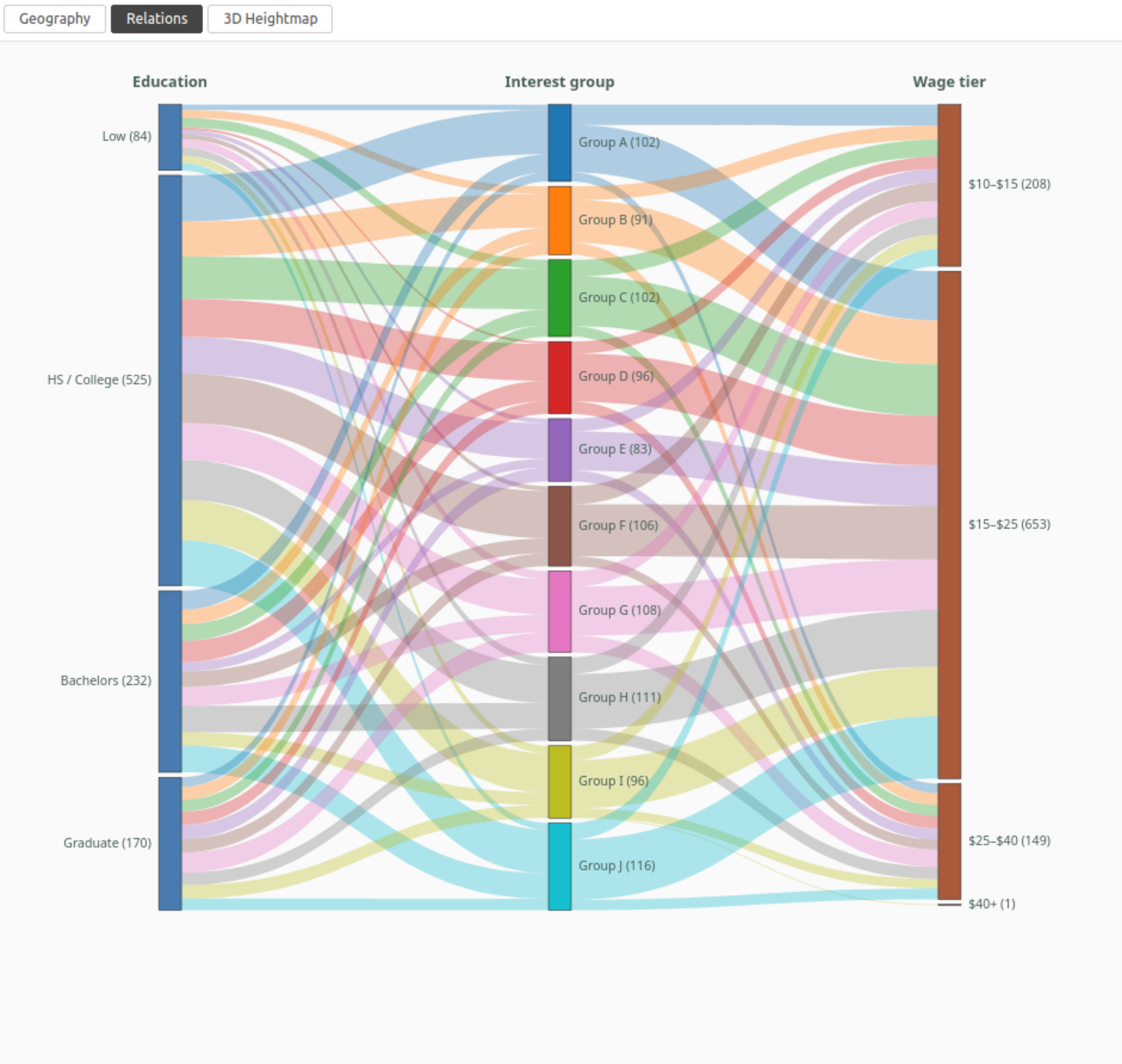

Figure 12: Sankey: Education → Interest Group → Wage tier. Ribbons are broadly parallel; no group is dominated by a single education or wage band.

3 Q3 — Business Base of Engagement

The dataset contains 253 employers and 1,011 participants. We characterise the business base along four axes: industry composition, wage structure, employer scale, and the degree to which workplaces function as social hubs.

Wage premium tracks education, but with a wide overlap. The wage scatter by education requirement (Fig. 13) shows a clear ordinal premium: median wage rises monotonically across the four education tiers (Low → HS/College → Bachelors → Graduate). However, the inter-tier overlap is substantial. A high-paying HS/College role can out-earn a low-paying Bachelors role. This suggests the labour market rewards education on average without enforcing strict tier-based pay floors — consistent with a service-economy mix of skilled trades, clerical roles, and professional positions.

Employer scale: long tail of small employers. The employer-size distribution (Fig. 14) reveals a heavy concentration of small employers (1–3 jobs) and a long tail of medium-sized ones (4–9 jobs). No single employer dominates the labour market in absolute terms; the largest employer accounts for ~9 jobs out of ~1,011 employed participants. Engagement is structurally a town of small and medium-sized businesses rather than one or two anchor firms.

Workplace social density: workplaces are social hubs. Of all coworker-pair friendships available to form, 65% actually exist in the social network — compared to a baseline of 16% for random participant pairs city-wide. The top 10 employers by job count show 100% friendship-pair cohesion: every employee at these firms is connected to every coworker. This is the strongest non-demographic predictor of friendship in the dataset, and it explains why the geographic friendship arcs (Q2 Pattern 7) span the city — workplaces draw together residents from all neighbourhoods.

Schedule structure. ~78% of jobs run a conventional Monday–Friday schedule. The remaining ~22% include retail, hospitality, and service positions with weekend or evening hours. Schools (5 in total) cluster the Monday–Friday daytime mode with relatively low pay, consistent with public-education economics.

Per-employer detail. Selecting any employer dot on the map opens a per-employer detail panel (Fig. 15) that reports: job count, wage range and median, education-mix of the workforce, household composition of employees, mean commute distance (computed via Haversine on residential and workplace coordinates), and the share of coworkers who are mutual friends. This panel is the primary tool for investigating whether a given employer differs structurally from the city baseline (e.g. unusually concentrated education tier, atypically long commutes, or weak workplace cohesion).

Pubs and restaurants — venues, not employers. The dataset distinguishes restaurants, pubs, and workplaces as venue types. Pubs concentrate weekend visits (51% of all pub check-ins fall on Saturday or Sunday) while restaurants are flat across the week (30% weekend share). Two pub archetypes emerge: "destination" pubs draw ~837 unique visitors with moderate per-visitor frequency, while "neighbourhood" pubs draw ~314 unique visitors with high per-visitor frequency (100+ visits each). This finding straddles MC1/MC2 boundaries and is reported here only because the *venue role in the social fabric* is a structural property (MC1) even where the visit timing is temporal (MC2).

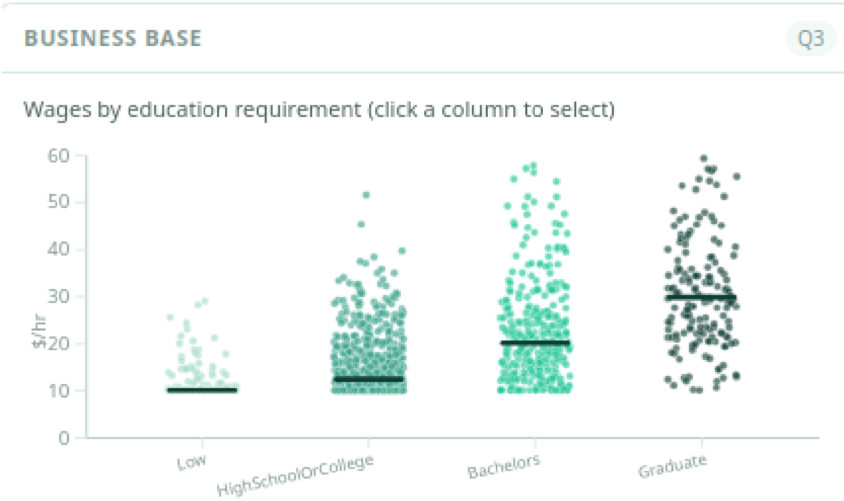

Figure 13: Wages grouped by education requirement. Median wage rises monotonically across tiers; inter-tier overlap is substantial.

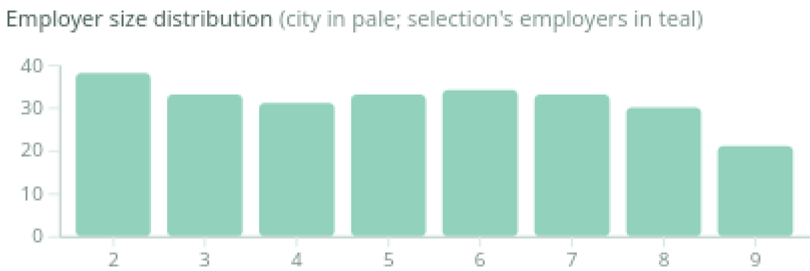

Figure 14: Employer-size distribution. Most employers offer 1–3 jobs; a long tail extends to 9 jobs.

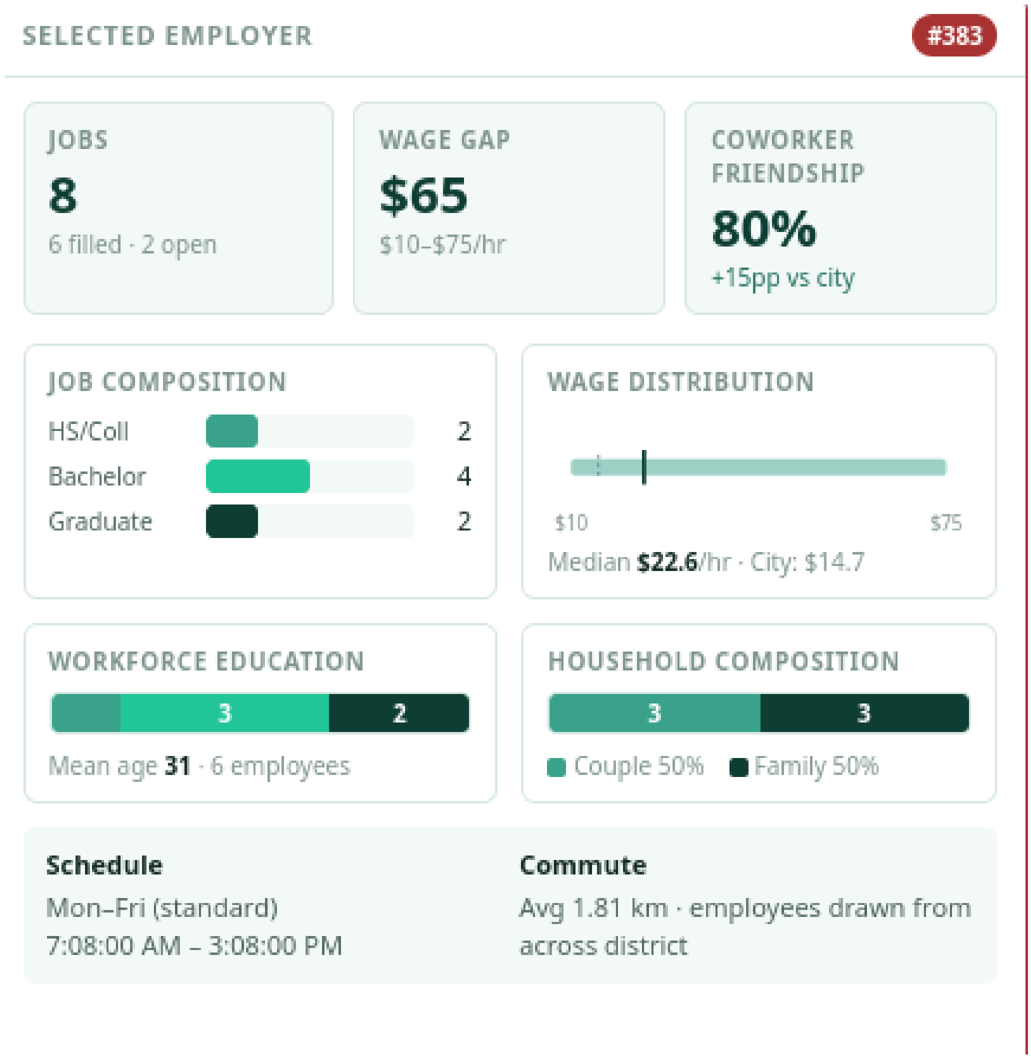

Figure 15: Per-employer detail panel: KPIs, wage range, education mix, household composition, schedule mode, commute distance, and coworker friendship density for a selected employer.

4 Q4 — One-Page Summary for Residents

Audience: Engagement residents (non-technical).

Engagement is a balanced, mid-sized town. The 1,011 of you who took part in this study describe a community that is demographically even-keeled: ages spread evenly between 18 and 60, education levels well-distributed across the population, with a slight bump in graduate-degree holders among the 25-34 generation. Most of you have at least a high-school diploma, and roughly two in five hold a bachelor's degree or higher.

Your friendships matter more than your money. The strongest predictor of how happy you reported feeling was not your wage, your education, or even your interest group — it was how many friends you have. Residents with 200+ social connections were overwhelmingly the happiest; those with under 50 connections clustered in the low-happiness range. The town's interest groups (A through J) do not differ much by age, income, or education, but they do differ clearly in social connectivity. Group B residents have on average 396 unique connections; Group H has 290.

Workplaces are where many friendships form. 65% of all coworker pairs in Engagement are friends in the social network — compared to just 16% for randomly selected resident pairs. At the town's ten largest employers, every coworker is connected to every other coworker. Where you work shapes who you know.

Your social life isn't bounded by your neighbourhood. Whatever neighbourhood you live in, your friends likely live across the city. The friendship arcs we drew on the map span the full town, not local clusters. This is healthy: it means Engagement is a genuinely connected community, not a collection of socially isolated districts.

The town's economy is small businesses, not anchor firms. 253 employers serve the town. Most have 1–3 jobs; the largest has just 9. No single firm dominates. Wages rise with education on average, but the overlap between tiers is wide enough that a high-paying high-school-required role can out-earn a low-paying bachelor's-required one.

Where the town spends its money. Of the average resident's monthly outgoings, shelter ($680) is the largest expense. Food ($345) and recreation ($398) are roughly matched, and education spending is minimal. Median rent is $707 — broadly consistent with the shelter figure, indicating low ownership.

Two findings the town might reflect on. First, Group H is the largest interest group, has above-average income, and is the most socially active by raw interaction count — yet reports the lowest joviality. High activity volume does not translate to wellbeing. Second, smaller, tighter groups (notably Group E at 83 members) report the highest happiness despite average or below-average earnings. Tight community appears to matter more than individual prosperity.

5 Methodology and Caveats

5.1 Scope discipline (MC1 only)

This work addresses MC1 (Demographics and Relationships) exclusively. The `ParticipantStatusLogs` (~5 GB across 72 files of 5-minute snapshots) are the primary data source for MC2 (Patterns of Life) and were therefore not ingested. One bounded exception: 3 of 72 status-log files were processed once to compute cross-sectional financial summaries (mean balance, food and weekly budget) per participant — covering 87% of participants from a subset of the time period. No temporal or trajectory analysis was performed on these data.

5.2 The vanishing cohort

Of 1,011 participants, 131 disappear from the data after month one and never return. This subset is not random: their mean joviality is 0.681 versus 0.466 for the retained group; 0% live alone; 41% have children; and all are in the Low or HighSchoolOrCollege education tiers. This is a survivorship bias affecting all post-month-1 findings and is acknowledged in the per-figure interpretations where applicable. Where we use lifetime aggregates, the bias is bounded; for short-window analyses it would be more severe.

5.3 Confidence threshold for building statistics

Residential buildings with fewer than 3 sampled residents are de-emphasised in map views via reduced fill opacity (0.55) and diagonal hatching, signalling that per-resident means are unreliable at small n. The threshold is chosen to balance noise reduction against information loss: 73% of residential buildings retain full opacity.

5.4 Colour choices

Joviality is rendered with a sequential `plasma` colormap fitted to the empirical range $[0.42, 0.56]$ rather than the theoretical $[0, 1]$. We deliberately avoid divergent palettes (e.g. `RdBu`) for joviality because 0.5 has no semantic meaning and the white midpoint of common divergent palettes collides with the "no data" cartographic convention. Categorical interest-group colours are drawn from a perceptually-uniform 10-class palette; the friendship-arc gradient (hot pink → vivid orange) is intentionally outside the spine teal/emerald palette to avoid collision with map encoding.

5.5 Selection and coordination

All four dashboard panels and the map share a single selection store. A selection is a set of `participantId`s annotated with a source (chart/building/employer/region). Cross-panel coordination is non-modal: every interaction flows through the same store, every view re-renders consistently. The 2×2 dashboard grid keeps all four panels visible simultaneously, supporting comparative exploration without tab-switching.

6 Tools, LLM Disclosure, and Effort

6.1 Tool stack

- **Backend**: Node.js + TypeScript, DuckDB (with native Parquet I/O), Express. The ingestion pipeline converts the raw CSVs to columnar Parquet, and SQL aggregation produces ~25 derived JSON files consumed by the frontend.
- **Frontend**: Vite + TypeScript, D3.js for 2D visualisation, Three.js for the 3D heightmap (custom shader for animated arcs).
- **Repository**: github.com/Redabelcaid/HPCDA-Project-Demographics-Relationships. Reproducible via `npm install` (backend and frontend) and `npm run dev` in two terminals; data is committed as Parquet + DuckDB so reviewers do not need to re-derive.

6.2 LLM disclosure

Large language models were used during development for: (i) drafting and reviewing TypeScript/SQL code, (ii) discussing visualisation design choices and tradeoffs, (iii) assisting with the prose drafts of this report. All analytical decisions, dataset scoping, methodological caveats, and findings interpretation were made by the human authors. The statistical analysis (Kruskal-Wallis, chi-square, eta-squared) was conducted manually using `scipy.stats`; results are reported verbatim from those computations.

6.3 Effort

Approximately 120 hours of cumulative team time across approximately 5 weeks: ~80 hours on the visual analytics tool (backend pipeline, frontend views, 3D heightmap, coordinated dashboard, lasso interaction, animated arcs); ~30 hours on data analysis (statistical tests, group profiles, social network characterisation); ~10 hours on report drafting, figure preparation, and reproducibility setup.

6.4 Video

A four-minute walkthrough of the tool is available at: insert video URL before submission.